Suggested Solutions to 2016 H2 A Level Examination Paper 3

1(a)	Electron's gain in k.e. = loss in e.p.e. $\frac{1}{2}mv^2 = qV$ $v = \sqrt{\frac{2qV}{m}}$ $= \sqrt{\frac{2(1.6\times10^{-19})(850)}{(9.11\times10^{-31})}}$ $= 1.7\times10^{7}$ m s^{-1}
1(b)(i)	Time to pass through the plates$= \frac{0.051}{1.7\times10^{7}}$ $= 3.0\times10^{-9}$ s Acceleration in vertical direction = $\frac{4.0\times10^{-15}}{9.11\times10^{-31}}$ $= 4.39\times10^{15}$ m s^{-2} $v_y = u_y + a_y t$ $= 0 + (4.39\times10^{15})(3.0\times10^{-9})$ $= 1.3\times10^{7}$ m s^{-1}
1(b)(ii)	v = $\sqrt{(1.7\times10^{7})^2 + (1.3\times10^{7})^2}$ $= 2.1\times10^{7}$ m s^{-1}
2(a)	Evaporation occurs at any temperature while boiling occurs at boiling point only